

Exercises - IV

- (1) Consider two 1-forms $\alpha_0 = dz + r^2 d\varphi$ and $\alpha_1 = \cos r dz + r \sin r d\varphi$ on $\mathbb{R}^3$, where (r, φ, z) denotes the cylindrical coordinate system on $\mathbb{R}^3$.
 - (a) Prove that α_0 and α_1 are contact forms on $\mathbb{R}^3$. (Recall that the volume form on $\mathbb{R}^3 \setminus \{(0, 0, 0)\}$ is given by $r dr \wedge d\varphi \wedge dz$.)
 - (b) Show that the contact forms α_0 and α_1 are invariant under the following diffeomorphisms:
 - (i) $f(r, \varphi, z) = (r, \varphi, z + c)$, where c is any constant.
 - (ii) $g(r, \varphi, z) = (r, \varphi + d, z)$, where d is any constant.
- (2) Note that S^n is the boundary of the unit ball in $\mathbb{R}^{n+1}$. Orient S^n by the outward normal radial vector field. [That is, for $p \in S^n$, an ordered basis $v_1, v_2, \dots, v_n$ of the tangent space $T_p S^n$ is positively oriented if and only if $p, v_1, \dots, v_n$ is a positively oriented basis of $\mathbb{R}^{n+1}$.]
 - (a) Prove that the antipodal map $a : S^n \rightarrow S^n$ is orientation preserving if and only if n is odd.
 - (b) Prove that $\mathbb{R}P^n$ is oriented if and only if n is odd.
 - (c) Recall that S^{2n+1} is a contact manifold. Define a contact form on $\mathbb{R}P^{2n+1}$.
- (3) Let T^*X denote the cotangent bundle of a smooth manifold X . Consider the sphere bundle $S(T^*X) \subset T^*X$. Show that the tautological form λ on T^*X restricts to a contact form on $S(T^*X)$. (Hint: Recall the description of λ .)